The Quantum Fourier Transform, Phase Estimation, and Shor's Algorithm

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Math 481 — Complex Analysis

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Roadmap

DFT $\rightarrow$ QFT $\rightarrow$ QPE $\rightarrow$ Shor's

Goal: factor a large integer N in time polynomial in $\log N$.

The DFT: a change of basis from roots of unity

$$\hat{x}_k = \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x_l \, \omega^{kl}, \qquad \omega = e^{-2\pi i/N}$$

Orthogonality of roots of unity

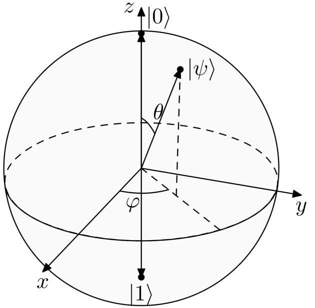
$$\frac{1}{N} \sum_{k=0}^{N-1} \omega^{mk} = \begin{cases} 1 & N \mid m \\ 0 & \text{else} \end{cases}$$

Proof for $N \nmid m$: with $r = \omega^m \neq 1$, geometric series gives $\frac{1}{N} \cdot \frac{r^N - 1}{r-1} = \frac{1}{N} \cdot \frac{e^{-2\pi i m} - 1}{r-1} = 0$.

Qubits

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1$$

$$n \text{ qubits} \;\longleftrightarrow\; \mathbb{C}^{2^n} = (\mathbb{C}^2)^{\otimes n}$$



The gate set

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}$$

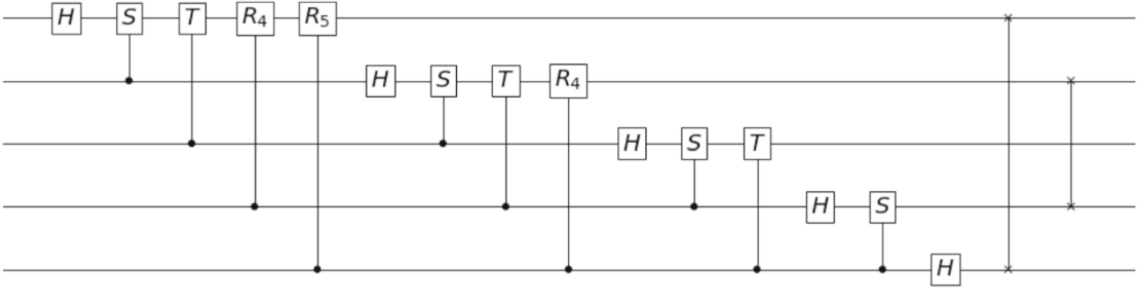
$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad C(R_k) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{2\pi i/2^k} \end{bmatrix}$$

QFT: the DFT factorizes over qubits

$$F_{2^n}|j_1 \cdots j_n\rangle = \frac{1}{\sqrt{2^n}} \left(|0\rangle + e^{2\pi i\, 0.j_n}|1\rangle\right) \otimes \cdots \otimes \left(|0\rangle + e^{2\pi i\, 0.j_1 \cdots j_n}|1\rangle\right)$$

One qubit per factor $\Rightarrow$ $O(n^2)$ gates instead of $O(2^n)$.

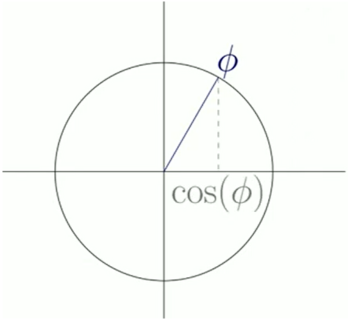
QFT circuit



QPE: eigenvalues of unitaries are phases

If $U|\psi\rangle = \lambda|\psi\rangle$ with U unitary, then

$$|\lambda| = 1, \qquad \lambda = e^{2\pi i\theta}, \quad \theta \in [0,1).$$



Proof

$\||\psi\rangle\| = \|U|\psi\rangle\| = |\lambda|\||\psi\rangle\| \Rightarrow |\lambda| = 1.$

Goal of QPE: find θ.

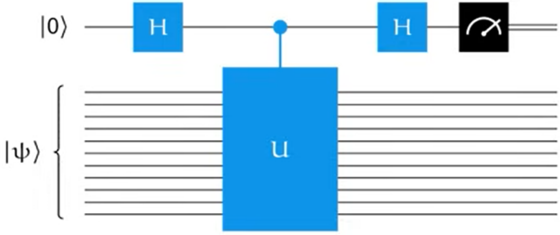
Phase kickback

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle \xrightarrow{C(U)} \frac{1}{\sqrt{2}}\big(|0\rangle + e^{2\pi i\theta}|1\rangle\big) \otimes |\psi\rangle$$

When we do a controlled gate, controlled by a qubit that has been put into superposition by a Hadamard gate, the control basically gets "reversed."

The target is unchanged, the phase lands on the *control*.

One ancilla: estimate θ by sampling



$$P(0) = \cos^2(\pi\theta), \qquad P(1) = \sin^2(\pi\theta)$$

One ancilla: where $\cos^2(\pi\theta)$ comes from

After kickback, apply the second Hadamard to $\frac{1}{\sqrt{2}}\big(|0\rangle + e^{2\pi i\theta}|1\rangle\big)$:

$$H|0\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\tfrac{1}{\sqrt{2}}\big(|0\rangle + e^{2\pi i\theta}|1\rangle\big) \xrightarrow{H} \tfrac{1}{2}\big(1 + e^{2\pi i\theta}\big)|0\rangle + \tfrac{1}{2}\big(1 - e^{2\pi i\theta}\big)|1\rangle$$

Born rule, using $|1 \pm e^{2\pi i\theta}|^2 = 2 \pm 2\cos(2\pi\theta)$:

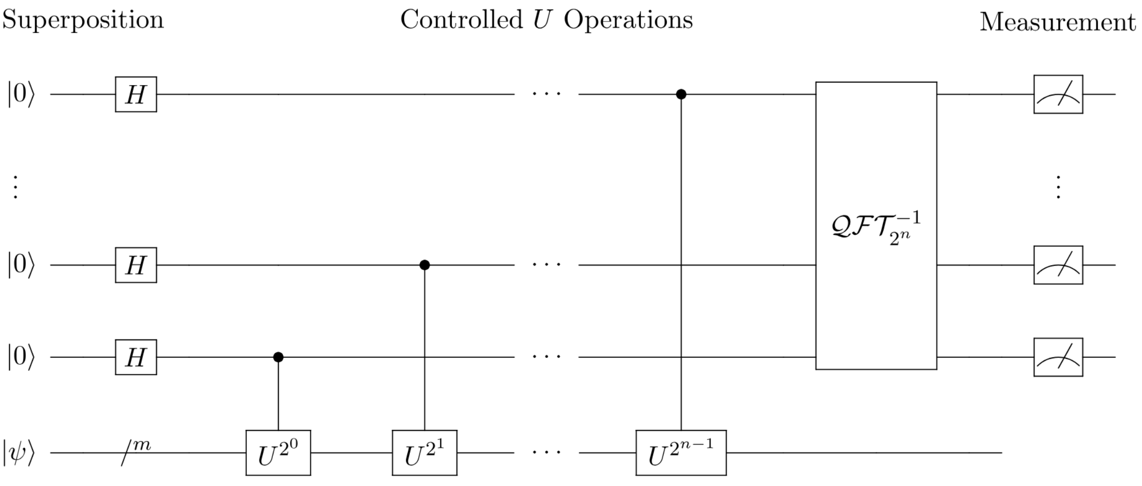
$$P(0) = \tfrac{1}{4}\big|1 + e^{2\pi i\theta}\big|^2 = \cos^2(\pi\theta), \qquad P(1) = \tfrac{1}{4}\big|1 - e^{2\pi i\theta}\big|^2 = \sin^2(\pi\theta)$$

Many ancillas: the kickbacks applied to the ancilla are effectively a QFT

$$\frac{1}{\sqrt{2^t}}\left(|0\rangle + e^{2\pi i\, 2^{t-1}\theta}|1\rangle\right) \otimes \cdots \otimes \left(|0\rangle + e^{2\pi i\, \theta}|1\rangle\right) \;=\; F_{2^t}|\tilde{\theta}\rangle$$

Apply $F_{2^t}^{-1}$, measure: θ in binary, one shot.

QPE circuit


Superposition
Controlled U Operations
Measurement
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|\psi\rangle$
H
H
H
m
U^{2^0}
U^{2^1}
$U^{2^{n-1}}$
$\mathcal{QFT}_{2^n}^{-1}$

Shor's: the number theory

Order of a: smallest r with $a^r \equiv 1 \pmod{N}$. Let $b = a^{r/2}$ (r even):

$$(b-1)(b+1) = b^2 - 1 \equiv 0 \pmod{N}$$

If $b \not\equiv \pm 1$: N divides the product but neither factor $\Rightarrow$ its primes split between them

$$\Rightarrow \ \gcd(b-1,\ N) \text{ is a nontrivial factor of } N.$$

Shor's: order-finding is QPE

$$U|x\rangle = |ax \bmod N\rangle \qquad \Longrightarrow \qquad \text{eigenvalues } e^{2\pi i\, s/r}$$

QPE measures $\phi \approx s/r \;\to$ continued fractions give $r \;\to \gcd(a^{r/2} \pm 1,\ N)$.

Example: $N = 15$, $a = 7$: $r = 4$, $7^2 \equiv 4$, $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$.

Takeaways

roots of unity $\rightarrow$ QFT $\rightarrow$ QPE $\rightarrow$ Shor's

- QFT: amplitudes $\rightarrow$ phases, in $O(n^2)$ gates.
- QPE: kick an eigenphase onto ancillas, decode with F^{-1}.
- Shor's: factoring $=$ order-finding $=$ QPE on $|x\rangle \mapsto |ax \bmod N\rangle$.